

FREE PARTICLE

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + 0 \right] \psi(x) = E \psi(x)$$

For $-\infty < x < +\infty$

$$\Rightarrow -\frac{d^2 \psi(x)}{dx^2} = \frac{2mE}{\hbar^2} \psi(x)$$

$$\Rightarrow \frac{d^2 \psi(x)}{dx^2} + \underbrace{\frac{2mE}{\hbar^2}}_{k^2} \psi(x) = 0$$

$$\Rightarrow \frac{d^2 \psi(x)}{dx^2} + k^2 \psi(x) = 0 \dots \dots \textcircled{1}$$

NOTE

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0$$

$$k^2 = \frac{2mE}{\hbar^2} \Rightarrow \boxed{\begin{array}{l} \text{Energy} \\ E = \frac{\hbar^2 k^2}{2m} \end{array}}$$

For k of real values
including ZERO!

As k can have real values, Energy is nothing but Continuous!

Free particle is not a quantum particle!

General
Solution of

Eq. 1

$$\Psi(x) = A e^{ikx} + B e^{-ikx}$$

A, B :
arbitrary
constants

$B=0 // \Psi(x) = A e^{ikx}$

$A=0 // \Psi(x) = B e^{-ikx}$

Euler's Formula :

$$e^{\pm i k x} = \cos kx \pm i \sin kx$$

Transferring exponential $\rightarrow$ Cosine + Sine form

$$\psi(x) = A e^{i k x} + B e^{-i k x}$$

$$= A (\cos kx + i \sin kx) + B (\cos kx - i \sin kx)$$

$$= \underbrace{(A+B)}_C \cos kx + i \underbrace{(A-B)}_D \sin kx$$

$$\Rightarrow \boxed{\psi(x) = C \cos kx + D \sin kx} \quad | \quad C, D : \text{constants}$$

Probability Density : $P(x) = |\psi(x)|^2$

$$= \psi^*(x) \psi(x)$$

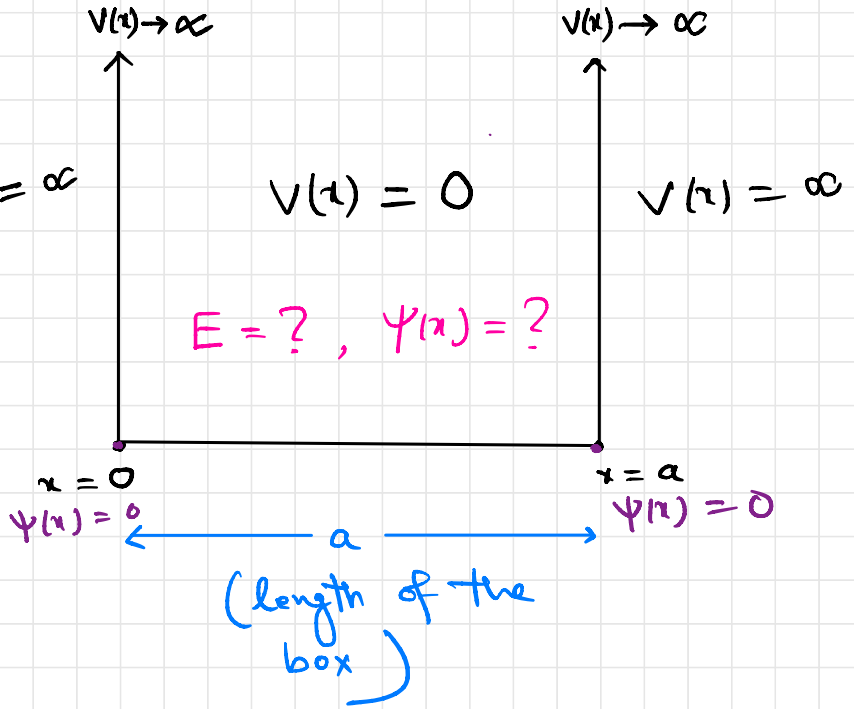
Probability
of finding
a particle
in region of space

$$: \int_{x_1}^{x_2} P(x) dx = \int_{x_1}^{x_2} \psi^*(x) \psi(x) dx$$

$$\boxed{\text{Ex. 1 : } \psi(x) = e^{-i k x} \\ \psi^*(x) = e^{+i k x}}$$

PARTICLE in ONE-DIMENSIONAL BOX

Constraining the motion of free particle $\rightarrow$ achieving energy quantization!



Boundary Conditions

$$\begin{aligned} \psi(x) &= 0 & \text{for } x=0 \\ \psi(x) &= 0 & \text{for } x=a \\ \psi(x) &\neq 0 & 0 < x < a \end{aligned}$$

Wavefunctions

Energy obtained from free particle model, so for

$$\psi(x) = C \cos kx + D \sin kx$$

[C, D : Constants]

$$E = \frac{k^2 \hbar^2}{2m}$$

[For all k real values]